# Unit 2, Lesson 5: Conditionals and Biconditionals

## Benchmark

MA.912.LT.4.3 Identify and accurately interpret "if…then," "if and only if," "all" and "not" statements. Find the converse, inverse, and contrapositive of a statement.

## Learning Targets

- ☐ I can write the converse, inverse, and contrapositive of a mathematical statement.
- ☐ I can determine if a statement is biconditional.
- ☐ I can write a definition as a biconditional.

## 2.5.1 Warm-Up: Streaming Payouts

The payouts to a musician for each time a song is streamed on multiple streaming platforms in 2019 is shown.

| Platform | Payout Per Stream |
|---|---|
| Napster | \$0.01900 |
| Tidal | \$0.01250 |
| Apple Music | \$0.00735 |
| Google Play | \$0.00676 |
| Deezer | \$0.00640 |
| Spotify | \$0.00437 |
| Amazon | \$0.00402 |
| Pandora | \$0.00133 |
| YouTube | \$0.00069 |

1. Determine the payout for a musician whose song was streamed 100,000 times on each platform.

| Platform | Payout for 100,000 Streams (in dollars) |
|---|---|
| Napster | |
| Tidal | |
| Apple Music | |
| Google Play | |
| Deezer | |
| Spotify | |
| Amazon | |
| Pandora | |
| YouTube | |

2. How many streams on Spotify did it take to for a musician to make \$1 in 2019?

## 2.5.2 Activity: True or False

For each conditional statement given, consider whether its converse, inverse, and contrapositive statements are true or false. Record the results in the table for each conditional.

1. **Conditional**: If all of the angles of a triangle are less than $90°$, then the triangle is an acute triangle.

| Relationship to Conditional | True or False |
|---|---|
| Converse | |
| Inverse | |
| Contrapositive | |

2. **Conditional**: If the sum of two angles is $90°$, then the angles are complementary.

| Relationship to Conditional | True or False |
|---|---|
| Converse | |
| Inverse | |
| Contrapositive | |

3. **Conditional**: If a polygon has six sides, then it is a hexagon.

| Relationship to Conditional | True or False |
|---|---|
| Converse | |
| Inverse | |
| Contrapositive | |

4. **Conditional**: If a line or ray divides an angle into two equal angles, then the line or ray is the bisector of the angle.

| Relationship to Conditional | True or False |
|---|---|
| Converse | |
| Inverse | |
| Contrapositive | |

5. **Conditional**: If lines that lie on the same plane do not intersect, then the lines are parallel.

| Relationship to Conditional | True or False |
|---|---|
| Converse | |
| Inverse | |
| Contrapositive | |

6. **Conditional**: If the opposite sides of a quadrilateral are parallel, then the quadrilateral is a parallelogram.

| Relationship to Conditional | True or False |
|---|---|
| Converse | |
| Inverse | |
| Contrapositive | |

7. **Conditional**: If $\overrightarrow{XY}$ is in the interior of $\angle WXZ$, then $m\angle WXY + m\angle YXZ = m\angle WXZ$.

| Relationship to Conditional | True or False |
|---|---|
| Converse | |
| Inverse | |
| Contrapositive | |

8. **Conditional**: If two planes intersect, then their intersection is exactly one line.

| Relationship to Conditional | True or False |
|---|---|
| Converse | |
| Inverse | |
| Contrapositive | |

9. **Conditional**: If two lines have the same slope, then they are parallel.

| Relationship to Conditional | True or False |
|---|---|
| Converse | |
| Inverse | |
| Contrapositive | |

## 2.5.3 Bring It Together: Biconditionals

1. How many of the conditionals from the previous Activity were considered true for all of the statements — converse, inverse, and contrapositive?

2. For how many conditionals was the converse of the conditional considered true?

A conditional statement is considered ***biconditional*** when both the statement and its converse are true. In other words, if both directions of the statement are true – "If $P$, then $Q$" and "If $Q$, then $P$" – the statement is biconditional.

Biconditional statements can be written in the format of "**if and only if**."

A **definition** is an exact statement, written formally, of the meaning or description of a word. Even though definitions in mathematics are often written as conditional statements, all definitions are actually biconditional statements.

3. Write each conditional using "if and only if."

   a. If the sum of two angles is $90°$, then the angles are complementary.

   b. If the opposite sides of a quadrilateral are parallel, then the quadrilateral is a parallelogram.

## 2.5.4 Your Turn!

1. Complete the table for the conditional statement.

If all four angles at the point of intersection of two lines are right angles, then the lines are perpendicular.

| Relationship to Conditional | Statement | True or False |
|---|---|---|
| Converse | | |
| Inverse | | |
| Contrapositive | | |

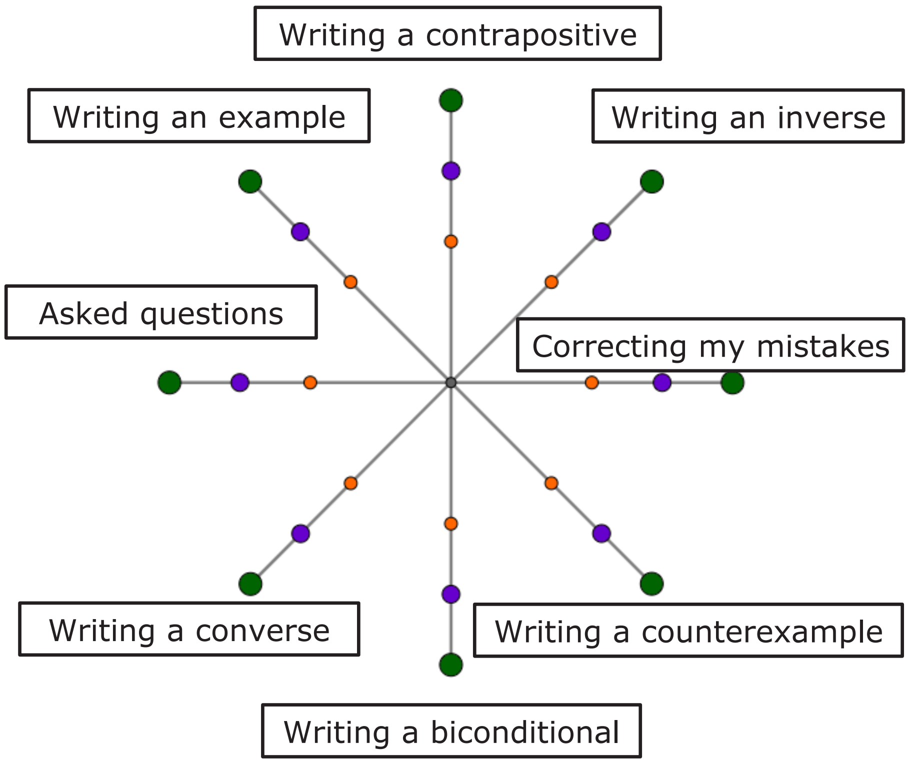
2. Explain why the conditional statement is biconditional.

3. Write the conditional using "if and only if."

## 2.5.5 Wrap-Up: Reflection

1. Place an **X** on the circle that shows what you learned or completed during today's lesson.

| | | | |
|---|---|---|---|
| | I am confident I know how to do this! | OR | I always did this! |
| | I am fairly sure I know how to do this. | OR | I tried this. |
| | I need support on this. | OR | I had trouble with this. |




# Unit 2, Lesson 6: "All" and "Not" Statements

### Benchmark

MA.912.LT.4.3 Identify and accurately interpret "if…then," "if and only if," "all" and "not" statements. Find the converse, inverse, and contrapositive of a statement.

### Learning Targets

- ☐ I can write and interpret an "all" statement.
- ☐ I can write and interpret a negation.